

ES 116 Obstacle Avoiding Robot Car Report

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Abstract– This lab report documents the design and implementation of an obstacle-avoiding robot car capable of autonomous navigation through a predefined path. The robot utilises an ultrasonic sensor to detect obstacles in its vicinity, initiating a response mechanism to halt forward motion upon detection. Subsequently, the robot analyses its surroundings to identify alternative paths, selecting the one with the least resistance to resume movement.

Introduction

This project aims to design and construct an obstacle-avoiding robot car capable of autonomously navigating through a predefined path. This robot is equipped with sensors to detect obstacles in its vicinity. When an obstacle is detected directly in front of the robot, it will initiate a response mechanism to halt its forward motion. Subsequently, the robot will analyse its surroundings to identify alternative paths available for traversal. By evaluating each potential path's resistance or obstruction level, the robot will determine the path with the least resistance, enabling it to resume movement along the optimal route. This obstacle avoidance and path selection mechanism is essential for the robot to navigate safely and efficiently in dynamic environments. Through this project, we aim to demonstrate the practical application of robotics in addressing real-world challenges such as autonomous navigation and obstacle avoidance.

Principle

The ultrasonic sensor utilised in this project operates on a fundamental principle: the transmitter emits ultrasonic waves, which bounce off any object close to proximity. The sensor's receiver then captures these reflected waves. By applying the formula $\text{speed} = \text{distance}/\text{time}$, we can calculate the distance to the object directly in front of the car. Should this distance fall below a predetermined threshold, the system instructs the motor driver to halt forward motion and initiate a slight reverse movement.

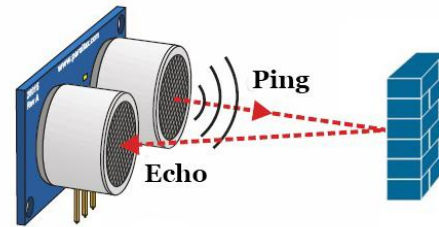


Fig.1 Working of Ultrasonic Sensor [1]

Following this, the servo motor undergoes rotation spanning from -90 degrees to 90 degrees. This rotation enables the ultrasonic sensor to scan across its full range, facilitating comprehensive detection of potential obstacles. Once the least obstructed pathway is identified, the car advances along this route until it encounters any subsequent obstructions. Should additional obstacles arise, the car repeats the aforementioned process. Through this iterative procedure, the car autonomously navigates along a path devoid of obstacles, thereby achieving automated operation.

Instrument Required

1. Ultrasonic Sensor (HC-SR04)
2. Arduino UNO
3. Servo Motor
4. Bread Board
5. L293D Motor Driving Sensor
6. Switch
7. Four DC Motors
8. Battery

Procedure

The vehicle progresses forward in a straight trajectory until it encounters an obstacle detected by an ultrasonic sensor. Upon detection, the vehicle reverses its direction, with coordination facilitated by the L293D motor driver sensor interfaced with the Arduino board. Pin connections on the Arduino are secured via soldering, ensuring stable communication. The L293D motor driver sensor regulates the operation of gear motors through its port connections.

Upon reversing, the ultrasonic sensor, mounted on a servo motor and linked to the designated servo motor pins on the Arduino, rotates a full 180 degrees. It gauges distances in all directions, identifying the path of least resistance. Subsequently, the vehicle adjusts its direction accordingly, facilitated by its three wheels, and resumes movement along the chosen path. This cycle perpetuates, enabling continuous operation of the vehicle.

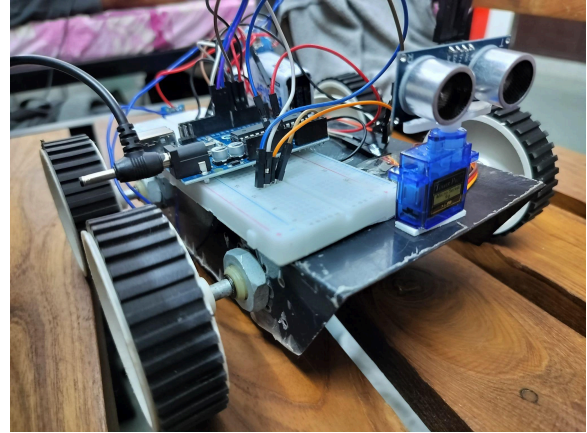


Fig.1 Final Picture of Project

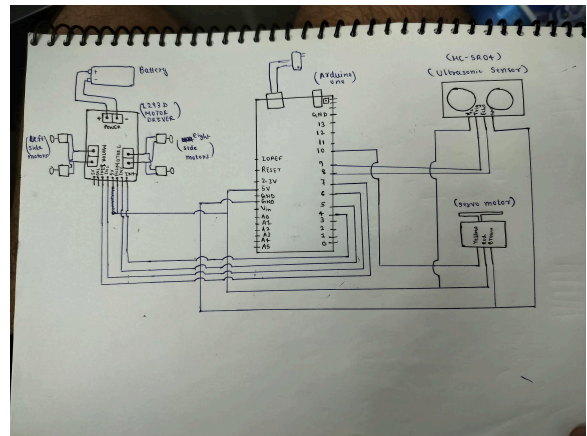


Fig.2 Circuit Diagram

Discussion

Our car was running really well, but it was facing issues at some points. Those issues are mentioned below:

1. Power Supply: Our main problem was giving our circuit a power supply. As per availability, we tried to use a 9v battery, but that did not last for a long time and got drained easily. We tried to solve this problem by using series and parallel connections, but that didn't work. Hence, we provided a power supply using a power adaptor directly to the Arduino.

2. Height Issue: The second problem we spotted during the trial was that if an object has a small height level compared to the height level of the ultrasonic sensor (HCSR04), the sensor cannot detect this object as it works on the principle of transmission and reflection signals.

Acknowledgement

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References

- [1] circuitgeeks, "Complete Guide for HC-SR04 Ultrasonic Sensor with Arduino," *Circuit Geeks*, Feb. 07, 2021.
<https://www.circuitgeeks.com/hc-sr04-ultrasonic-sensor-with-arduino/>